LP-DiD with inverse probability weighting and doubly robust estimation

A methodological note on the `lpdidcsa` R package*
draft – to be revised

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Abstract

This note shows how inverse probability weighting and doubly robust estimation can be combined with the Local Projection Difference-in-Differences (LP-DiD) estimator of Dube et al. (2025). We first present LP-CSA, a local-projection version of the Callaway and Sant'Anna (2021) estimator, which recovers all cohort-specific treatment effects in a single regression per horizon. We then present three ways of accounting for pretreatment covariates while targeting the equally weighted average treatment effect on the treated (ATT): regression adjustment, inverse probability weighting, and a doubly robust estimator that combines the two. The doubly robust estimator remains consistent if either the outcome model or the propensity score model is correctly specified. We also provide standard errors that account for the estimation of weights and first-stage models. All estimators are implemented in the `lpdidcsa` R package, which also handles missing observations through pair balancing, non-absorbing treatments, and high-dimensional fixed effects.

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1 Introduction

Difference-in-differences with staggered treatment adoption is now usually estimated with methods that avoid "forbidden comparisons" between newly treated and already treated units. Two families of estimators are popular. Callaway and Sant'Anna (2021) (CSA) compute one simple difference-in-differences for each treatment cohort and each horizon, and then average them. Dube et al. (2025) propose LP-DiD, which obtains comparable estimates with local projections: one regression per horizon, estimated on a "clean" sample that excludes already treated units.

This note makes three points.

1. LP-DiD's clean sample also allows estimating all CSA cohort-specific effects in a single regression per horizon. We call this regression *LP-CSA*. It gives the same average effect as the reweighted LP-DiD of Dube et al. (2025), but it also reports the effect of each cohort.
2. Pretreatment covariates can be introduced in LP-DiD in three ways that preserve the equally weighted ATT: regression adjustment (RA), inverse probability weighting (IPW), and a doubly robust (DR) combination of both. RA relies on a correct model for the outcome, IPW on a correct model for the treatment, and DR on at least one of the two.
3. Because weights and first-stage models are estimated, usual regression standard errors are not exactly right. We derive corrected standard errors for each method.

All methods are implemented in the `lpdidcsa` R package, built on `fixest` (Bergé, 2018). Section 2 sets the notation. Section 3 presents the estimators without covariates, and Section 4 the estimators with covariates. Section 5 discusses inference. Section 6 summarizes the main practical advantages of the LP framework, and Section 7 presents the package. The appendices detail the standard-error corrections.

2 Setting and notation

Units are indexed by i and periods by s. $T_{s,i}$ is a dummy equal to 1 when unit i is treated in period s. A unit *starts* its treatment in t when $\Delta T_{t,i} = T_{t,i} - T_{t-1,i} = 1$. Units starting their treatment in t form *cohort* t. We study the effect of the treatment h periods after it starts (horizon h), always relative to the last period before treatment, $t-1$.

Long differences. All the estimators below use the long difference of the outcome between $t+h$ and $t-1$:

$$\Delta^h y_{t,i} = y_{t+h,i} - y_{t-1,i}. \tag{1}$$

For $h = 0$, this is the usual first difference $y_{t,i} - y_{t-1,i}$. For other horizons, it is a long (forward or backward) difference.[1]

Clean sample. For each horizon h, following Dube et al. (2025), the estimation sample contains:

- *treated observations*: units starting their treatment in t ($\Delta T_{t,i} = 1$);
- *clean controls*: units not yet treated in $t + \max(h, 0)$ ($T_{t+\max(h,0),i} = 0$), including never-treated units.

Already treated units are excluded, which rules out forbidden comparisons. A unit enters the sample of horizon h for date t only if it is observed both in $t-1$ and in $t+h$. We denote by n_t the number of treated observations of cohort t in this sample, and by q_t their share among all observations of date t.

The CSA estimator. For cohort t and horizon h, Callaway and Sant'Anna (2021) compute a simple difference-in-differences between the treated (1) and the clean controls (0):

$$\beta_t^h = \left(\overline{Y_{t+h}(1)} - \overline{Y_{t-1}(1)}\right) - \left(\overline{Y_{t+h}(0)} - \overline{Y_{t-1}(0)}\right), \tag{2}$$

and then average these cohort effects with weights proportional to cohort sizes:

$$\beta_{CSA}^h = \frac{\sum_t n_t \beta_t^h}{\sum_s n_s}. \tag{3}$$

We call β_{CSA}^h the *equally weighted ATT*: each treated observation counts equally.

3 Estimation without covariates

3.1 LP-CSA: cohort-specific effects in one regression

In a balanced panel, the mean of a difference equals the difference of the means. The cohort effect β_t^h is thus simply the difference between the average

[1] Pre-treatment horizons $h \leq -2$ are used to test for pre-trends. Horizon $h = -1$ is the reference period and is omitted.

long difference of the treated and that of the clean controls:

$$\beta_t^h = \overline{\Delta^h y_{t,i}(1)} - \overline{\Delta^h y_{t,i}(0)}, \tag{4}$$

which can be estimated with a regression on the clean sample of date t: $\Delta^h y_{t,i} = \alpha_t^h + \beta_t^h \Delta T_{t,i} + u_i$.

Since the clean sample of horizon h stacks all dates t, all these cohort effects can be estimated in a *single* regression per horizon, by interacting the treatment dummy with date dummies:

$$\Delta^h y_{t,i} = \underbrace{\sum_s \beta_s^h \left(\mathbb{1}_{s=t} \times \Delta T_{t,i}\right)}_{\substack{\text{Treatment dummy} \\ \text{interacted with dates}}} + \underbrace{\delta_t^h}_{\text{Date FE}} + u_{t,i}^h. \tag{5}$$

We call this regression LP-CSA. The coefficient β_s^h is exactly the CSA effect of cohort s. The average effect β_{CSA}^h is then obtained with the aggregation of equation (3), as in Sun and Abraham (2021) (e.g., with the `aggregate` function of `fixest`).

The main interest of LP-CSA is that it reports *each cohort's effect* together with the average. This is useful to check whether the average hides very different cohort effects, for instance between early and late cohorts. When the sample is not too large, the regressions of all horizons can also be stacked into one regression, which gives standard errors that account for the correlation of errors across horizons.

3.2 LP-DiD and reweighted LP-DiD

Dube et al. (2025) use the same clean sample, but with a single treatment dummy:

$$\Delta^h y_{t,i} = \underbrace{\beta^h \Delta T_{t,i}}_{\text{Treatment dummy}} + \underbrace{\delta_t^h}_{\text{Date FE}} + u_{t,i}^h. \tag{6}$$

This regression is faster than LP-CSA, because it has a single treatment coefficient. Its estimate is a weighted average of the cohort effects:

$$\beta_{LPDID}^h = \sum_t \omega_t \, \beta_t^h, \qquad \omega_t = \frac{N_t \, q_t(1-q_t)}{\sum_s N_s \, q_s(1-q_s)} = \frac{n_t(1-q_t)}{\sum_s n_s(1-q_s)}, \tag{7}$$

where N_t is the number of observations of date t in the clean sample, so that $N_t q_t = n_t$. The denominator only ensures that the weights sum to one. With date fixed effects, the regression compares treated and controls within each date, and weights each date by the variance of the treatment dummy, $N_t q_t(1-q_t)$. The weights ω_t are positive, but they differ from the CSA weights $n_t / \sum_s n_s$: cohorts with few clean controls (a high q_t, typically late cohorts)

receive less weight. Hence, in general, $\beta^h_{LPDID} \neq \beta^h_{CSA}$. Dube et al. (2025) call β^h_{LPDID} the *variance-weighted ATT*.

The fix is simple: weight each observation of date t by

$$w^{rw}_t = \frac{1}{1-q_t}. \tag{8}$$

Since this weight is the same for the treated and the controls of date t, it leaves q_t and β^h_t unchanged, and multiplies the weight of date t by w^{rw}_t. The factor $(1-q_t)$ then cancels out:

$$\omega^{rw}_t = \frac{w^{rw}_t\, n_t(1-q_t)}{\sum_s w^{rw}_s\, n_s(1-q_s)} = \frac{n_t}{\sum_s n_s}, \qquad \text{so that} \qquad \sum_t \omega^{rw}_t\, \beta^h_t = \beta^h_{CSA}. \tag{9}$$

In practice, $1-q_t$ is the residual, for treated observations, of a regression of $\Delta T_{t,i}$ on date fixed effects, which is how Dube et al. (2025) present it (any multiplicative constant in w^{rw}_t leaves the estimate unchanged).

LP-CSA or reweighted LP-DiD? Without covariates, reweighted LP-DiD and LP-CSA give *exactly the same* average effect. The difference is what they report: reweighted LP-DiD gives only the average, LP-CSA gives the average and every cohort-specific effect. Reweighted LP-DiD is faster; LP-CSA is more informative.

4 Estimation with covariates

Parallel trends may only hold conditionally on pretreatment covariates X_i: for instance, if men and women have different earnings trajectories and the treated are more often men. In what follows, X_i is measured before treatment (e.g., in $t-1$).

4.1 Linear controls

The simplest option adds the covariates linearly to equation (6) (or to (5)):

$$\Delta^h y_{t,i} = \beta^h \Delta T_{t,i} + \delta^h_t + \underbrace{\sum_k \psi^h_k X_{k,i}}_{\substack{\text{Pretreatment} \\ \text{control variables}}} + u^h_{t,i}. \tag{10}$$

This is very fast. Combined with reweighted LP-DiD or LP-CSA, it allows the treatment effect to vary across cohorts (and horizons), but not with the covariates. It relies on two restrictions. First, the treatment effect must not

vary with the covariates; otherwise, the estimate is again a variance-weighted average (Dube et al., 2025). Second, the covariates must affect outcome trends in the same way in every period: the earnings gap between men and women, for instance, must evolve at a constant pace over time.

Reweighted LP-DiD and LP-CSA with linear controls. The two estimators weight cohorts differently once covariates are included. LP-CSA estimates one effect per cohort and then averages these effects with weights n_t: it targets the equally weighted ATT exactly. Reweighted LP-DiD applies the weights w_t^{rw} before estimation. These weights are designed for a regression with date fixed effects only. Once covariates are added, the regression removes, within each cohort, the part of the treatment that the covariates predict. The cohort weights then no longer exactly equal $n_t / \sum_s n_s$, and the estimate is only approximately the equally weighted ATT. The difference is usually small. It grows when the covariate imbalance between treated and controls differs across cohorts.

The three methods below relax the first restriction and target the equally weighted ATT. Section 4.5 shows how to relax the second.

4.2 Regression adjustment

Regression adjustment proceeds in two steps.

Step 1. Estimate an outcome model on the clean controls only:

$$\Delta^h y_{t,i} = \delta_t^h + \sum_k \psi_k^h X_{k,i} + e_{t,i}, \qquad \text{for clean controls } (\Delta T_{t,i} = 0). \tag{11}$$

Step 2. Use this model to predict, for each treated observation, the long difference it would have experienced without treatment, $\widehat{\Delta^h y_{t,i}}(0) = \hat{\delta}_t^h + \sum_k \hat{\psi}_k^h X_{k,i}$. The ATT is the average gap between the actual and the predicted long differences of the treated:

$$\hat{\beta}_{RA}^h = \frac{1}{\sum_t n_t} \sum_{i:\, \Delta T_{t,i}=1} \Big(\Delta^h y_{t,i} - \underbrace{\widehat{\Delta^h y_{t,i}}(0)}_{\substack{\text{predicted counterfactual} \\ \text{for the treated}}} \Big). \tag{12}$$

Because the average is taken over all treated observations, each cohort weighs in proportion to n_t: RA targets the equally weighted ATT. Without covariates, $\hat{\delta}_t^h$ is the average long difference of the clean controls of date t, and RA gives exactly β_{CSA}^h. This is the regression adjustment proposed by Dube et al. (2025). It is fast: two simple regressions per horizon. It is consistent if the outcome model (11) is correctly specified.

4.3 Inverse probability weighting

Inverse probability weighting, or propensity score weighting, is widely used in the DiD literature since Abadie (2005). Instead of modelling the outcome, it models the treatment. It reweights clean controls so that they resemble the treated in terms of covariates.

We first estimate, on the clean sample of horizon h, the probability of starting treatment with a logistic regression:

$$\text{logit}\left(P\left(\Delta T_{t,i}=1\right)\right) = \underbrace{\delta_t}_{\text{Date FE}} + \underbrace{\sum_k \psi_k X_{k,i}}_{\substack{\text{Pretreatment} \\ \text{control variables}}} . \tag{13}$$

This gives the weights:

$$w_{i,h}^{\text{ipw}} = \begin{cases} 1, & \text{if } \Delta T_{t,i}=1 \text{ (treated)}, \\ \dfrac{\hat{p}_i}{1-\hat{p}_i}, & \text{if } \Delta T_{t,i}=0 \text{ (clean control)}, \end{cases} \tag{14}$$

where $\hat{p}_i$ is the predicted probability of treatment. Controls that look like the treated (high $\hat{p}_i$) receive more weight.

Double reweighting. We then combine these weights with the reweighting of equation (8). Since the IPW weights differ between treated and controls, they change the treated shares: q_t is therefore recomputed with the weights w^{ipw}, which gives w_t^{rw}. The final weight is $w_{i,t,h}^{*} = w_{i,h}^{\text{ipw}} \times w_t^{rw}$, and it is used in the LP-DiD regression (6). The same weights w^{ipw} can be used in LP-CSA (equation 5) to obtain cohort-specific IPW effects. Numerical examples show that IPW LP-DiD and IPW LP-CSA give exactly the same average estimate.

Trimming. Controls with a propensity score close to 1 receive very large weights and can make the estimate unstable. By default, the package drops controls with $\hat{p}_i \geq 0.995$ (option `ps_trim`); treated observations are always kept.

Proposition: Re-weighted LP-DiD with inverse probability weighting enables estimating the "equal-weighted ATT"

Proof

Lemma (Row duplication). For any weighted least squares regression with non-negative rational weights $w_i \in \mathbb{Q}_+$, there exist a) an integer N such as $w_i \times N \in \mathbb{N}$ and b) an unweighted database DB$'$ constructed by duplicating

each observation i exactly $w_i \times N$ times, such that OLS on DB$'$ yields identical point estimates to WLS on DB.

Step 1. Dube et al. (2025) establish that on any unweighted clean long-difference database, re-weighted LP-DiD and LP-CSA yield the same equal-weighted ATT estimator β^h_{CSA}.

Step 2. Let DB be a clean long-difference database where each observation i carries a weight $w^{\text{ipw}}_{i,h}$ as defined in equation 13. By the Lemma, WLS on DB with weights w^{ipw} is equivalent to OLS on an unweighted database DB$'$, where each observation i is duplicated $w^{\text{ipw}}_{i,h}$ times.

Step 3. DB$'$ is an unweighted clean long-difference database. By Step 1, re-weighted LP-DiD and LP-CSA are equivalent on DB$'$, and both yield the equal-weighted ATT estimator β^h_{CSA}.

Step 4. Since WLS on DB with weights w^{ipw} is equivalent to OLS on DB$'$ (Step 2), and since LP-CSA and re-weighted LP-DiD are equivalent on DB$'$ (Step 3), it follows that LP-CSA and re-weighted LP-DiD with weights w^{ipw} on DB are equivalent. □

The proof shows that the two estimators coincide. That each cohort-specific IPW estimate identifies the ATT of its cohort follows from Abadie (2005), under conditional parallel trends and a correctly specified propensity score.

4.4 Doubly robust estimation

RA and IPW rely on different assumptions. RA is consistent if the outcome model is correct, IPW if the propensity score model is correct. In practice, we rarely know which model is better. Doubly robust estimators combine the two models and are consistent if *at least one* of them is correct (Sant'Anna and Zhao, 2020). This property explains their popularity in the DiD literature, and DR is the default method of the `did` package.

The DR LP-DiD estimator combines the two first steps above. It uses the outcome model of equation (11), estimated on the clean controls, and the propensity score model of equation (13), which gives the odds $\hat{w}_i = \hat{p}_i/(1-\hat{p}_i)$.

It then computes:

$$\hat{\beta}^h_{DR} = \overbrace{\frac{1}{\sum_t n_t} \sum_{i:\,\Delta T_{t,i}=1} \Big(\Delta^h y_{t,i} - \underbrace{\widehat{\Delta^h y_{t,i}(0)}}_{\substack{\text{predicted counterfactual} \\ \text{for the treated}}} \Big)}^{\text{RA estimate}} \\ - \overbrace{\sum_{i:\,\Delta T_{t,i}=0} \frac{\hat{w}_i}{\sum_{j:\,\Delta T_{t,j}=0} \hat{w}_j} \Big(\Delta^h y_{t,i} - \underbrace{\widehat{\Delta^h y_{t,i}(0)}}_{\substack{\text{fitted value} \\ \text{for the controls}}} \Big)}^{\text{IPW-weighted average of control residuals}} \tag{15}$$

Both lines use the same prediction $\widehat{\Delta^h y_{t,i}(0)}$, from the outcome model estimated on the controls. For the treated (first line), it is a predicted counterfactual, as in RA. For the controls (second line), it is the fitted value of the model, so that the term in parentheses is a residual. An unweighted average of these residuals would be zero, since the model is estimated on the controls with date fixed effects. Reweighting them to look like the treated tells whether the model fits as well for controls that resemble the treated.

- If the outcome model is correct, the control residuals average to zero, whatever the weights. The correction vanishes and DR is consistent, like RA.
- If the propensity score is correct, the reweighted controls resemble the treated. The correction then measures how much the outcome model is wrong for observations like the treated, and removes this error. DR is consistent, like IPW.

As with IPW, controls with $\hat{p}_i \geq 0.995$ are trimmed from the weighted term by default. As with RA, the average is taken over all treated observations, so DR targets the equally weighted ATT.

4.5 Interacting covariates with treatment cohorts

By default, the first-step models of RA, IPW and DR (equations 11 and 13) are estimated once per horizon, pooling all cohorts with date fixed effects. The effect of the covariates on the outcome, or on the probability of treatment, is then the same for all cohorts.

The `did` package instead estimates a separate model for each cohort and period. The option `time_x_controls = TRUE` does the same by interacting the

covariates with date dummies. For RA, the outcome model (11) becomes:

$$\Delta^h y_{t,i} = \delta_t^h + \sum_s \sum_k \psi_{s,k}^h \left(\mathbb{1}_{s=t} \times X_{k,i}\right) + e_{t,i}, \qquad \text{for clean controls,} \quad (16)$$

and for IPW, the propensity score (13) becomes:

$$\text{logit}\left(P\left(\Delta T_{t,i} = 1\right)\right) = \delta_t + \sum_s \sum_k \psi_{s,k} \left(\mathbb{1}_{s=t} \times X_{k,i}\right). \quad (17)$$

The coefficient of each covariate now differs by reference date. Since, at a given horizon, each reference date corresponds to one cohort, the effect of the covariates on outcome trends (RA) or on the probability of treatment (IPW) is allowed to differ across cohorts. DR combines the two interacted models. When high-dimensional fixed effects are included, they are also interacted with the dates. The same option interacts linear controls with dates in `lpdid`, `lpdid_rw` and `lpcsa`. With this option, and with not-yet-treated units as controls, `lpdidcsa` reproduces the estimates of `did` (cf. this vignette for a numerical example).

The package keeps the non-interacted models as default. They are faster, and interacted models can be unstable when some cohorts are small. In our applications, the two specifications give close estimates. Comparing them is nevertheless a useful robustness check.

5 Inference

All the estimators above involve quantities that are themselves estimated: the treated shares q_t used for reweighting, the propensity score, and the outcome model. A regression standard error that treats them as known ignores this first-stage uncertainty.

The package reports, for each horizon, the usual cluster-robust standard error (`se`) and, when requested, a corrected standard error (`se_correct`) that propagates the first-stage uncertainty. The correction follows the same logic for all methods. Each estimate can be written, to a first approximation, as a sum of contributions of individual observations, its *influence function*. The correction adds to the contribution of each observation its effect through the first-stage estimates. These contributions are then summed by cluster, which accounts for the fact that the same cluster can appear among both treated and controls.

- For RA and DR (`lpdid_adj`, `lpdid_dr`), the correction is computed by default (option `se_correction`). For RA, it reproduces exactly the standard error of the Stata `lpdid` command of Dube et al. (2025). For DR, it follows Sant'Anna and Zhao (2020).

- For reweighted LP-DiD and IPW LP-DiD (`lpdid_rw`, `lpdid_ipw`), the correction accounts for the estimation of the treated shares q_t and, for IPW, of the propensity score. It is optional, because it is usually small (option `se_correction_rw_ipw`).

The corrections use a single clustering variable, treat trimming as fixed, and do not propagate the uncertainty of absorbed high-dimensional fixed effects. Appendices A to C give the formulas.

6 Main advantages of the LP framework

Beyond speed, the LP framework has four practical advantages.

A simple regression representation. Every estimator is a regression, or two, on a clearly defined sample. Users can inspect each step with standard tools, see which observations are compared at each horizon, and add their own specifications. This makes the methods easy to teach and to check.

Missing observations and pair balancing. Each long difference only requires the unit to be observed in $t-1$ and $t+h$. The panel is thus balanced pair by pair (*pair balancing*), separately for each horizon, rather than for the whole period. This matters in panels of workers, where people regularly enter and exit the panel. Imposing a fully balanced panel may select very specific workers, misrepresent the phenomenon studied, and reduce statistical power. In R, the `did` package offers either a fully balanced panel or an unbalanced panel in which means are computed by group, which amounts to group rather than unit fixed effects. When the variance of unit fixed effects is large, the two can differ substantially. Estimates obtained on unbalanced and rebalanced panels are often much closer with the unit fixed effects implied by long differences (cf. this vignette for a numerical example).

Missing observations also raise a less visible problem: they can hide the start of a treatment. If a unit is untreated in 2003, unobserved in 2004 and treated in 2005, its treatment may have started in 2004 or 2005. The package lets users give the treatment start date directly (`last_treat`), so that it does not have to be inferred from observed rows. When only a treatment indicator is available, units whose start date is uncertain are excluded from both treated and controls over the uncertain interval. A unit observed in $t-1$ and $t+h$ but not in t still contributes to horizon h.

The LP format also makes alternative balancing rules easy to implement. For instance, one can require all units to be observed during a fixed window around

their treatment (e.g., 5 years before and 5 years after) rather than between two calendar dates. With staggered treatments, the calendar rule gives estimates at long horizons that mostly reflect early cohorts.

Non-absorbing treatments. Many treatments can happen several times: job losses, moves, promotions, births. With `absorbing = FALSE`, each new treatment start is a treated observation, and clean controls are units with no treatment start in the window, optionally after a minimum number of periods since their last treatment (option `reentry`). CSA's framework assumes that treatment, once started, never stops.

High-dimensional fixed effects. Since the estimation relies on `fixest`, high-dimensional fixed effects (e.g., firm or occupation) can be absorbed in all regressions, including the propensity score and outcome models. In a cell-by-cell approach, such controls would have to enter as dummies in many small samples. One limitation should be noted. When fixed effects are absorbed in the first-step models (the outcome model of RA and DR, or the propensity score), the standard-error corrections do not propagate the uncertainty of their estimation: only the coefficients of the explicit covariates and date dummies are taken into account. With absorbed fixed effects, the corrected standard errors are therefore only a partial correction.[2]

7 The `lpdidcsa` R package

The `lpdidcsa` R package implements all the methods discussed in this note.

7.1 Data preparation

The function `lpdidcsa_data` prepares the data. It identifies treatment starts (from a treatment indicator or, preferably, from the treatment start date `last_treat`), defines the variables needed to build clean samples, and computes the leads and lags of the outcome. Two formats are available. The *wide* format (default) adds one column per horizon. The *long* format stacks horizons in rows; it uses more memory but allows estimating all horizons in a single regression (`one_reg = TRUE`), with standard errors clustered across horizons.

[2] When the number of fixed-effect levels is moderate, they can instead be entered as explicit dummies in `controls` (e.g., `factor(firm)`), so that their uncertainty is fully propagated, at the cost of computation time.

Weights and clusters. Weights and clustering variables are read on the row of the treatment date t. When weights vary over time, they may already be affected by the treatment. We therefore recommend measuring them before treatment, in $t-1$, by creating their lagged value in the data step (e.g., `p_variables = list(tm1 = "emp")`) and using `weight = "emp_tm1"` in `lpdidcsa`. For clustering, the relevant level depends on the source of correlation: the pre-treatment firm captures shocks shared before treatment and firm-level treatment assignment, the firm in $t+h$ captures shocks shared after treatment (option `clusters_h`). Both can be combined with two-way clustering.

7.2 Estimation methods

The function `lpdidcsa` offers seven methods (Table 1).

Table 1: Methods available in `lpdidcsa`

Method	Covariates enter through	Average effect	Valid with covariates if ...
`lpdid`	linear controls	variance-weighted[a]	effects do not vary with covariates
`lpdid_rw`	linear controls	approximately equally weighted[b]	effects do not vary with covariates
`lpcsa`	linear controls	equally weighted[c]	effects do not vary with covariates
`lpdid_adj`	outcome model	equally weighted	the outcome model is correct
`lpdid_ipw`	propensity score	equally weighted	the propensity score is correct
`lpcsa_ipw`	propensity score	equally weighted[c]	the propensity score is correct
`lpdid_dr`	both	equally weighted	either model is correct

[a] Equal to the equally weighted ATT when the treatment effect is the same for all cohorts, since any weighted average of identical effects gives the same value.
[b] `lpdid_rw` reweights observations before controlling for covariates. If covariates predict treatment better in some cohorts, controlling for them partly undoes the reweighting.
[c] Also reports cohort-specific effects. Without covariates, `lpdid_ipw` reduces to `lpdid_rw`, `lpcsa_ipw` to `lpcsa`, and `lpdid_rw`, `lpcsa`, `lpdid_adj` and `lpdid_dr` all give the same estimate, β^h_{CSA}.

All methods accept survey weights (possibly varying by horizon), multiple clustering variables, absorbed fixed effects, anticipation periods, and non-absorbing treatments. The option `prediction = TRUE` also plots the average outcome path of the treated and its estimated counterfactual.

7.3 Execution time

To test the speed of the methods, we simulate a balanced panel of 5,000 individuals observed over 20 years (100,000 observations). Ten percent of individuals are treated, with treatment starts staggered between year 5 and year 15. We estimate all possible horizons, from $h = -14$ to $h = +15$. The control variable is a single dummy (gender). We then repeat the exercise on a large panel of 200,000 individuals (4,000,000 observations). The benchmark script is available here. We use a 64GB RAM Linux computer.

Tables 2, 3 and 4 report median execution times. We compare them with the `did` package, run on the same data with the same control variable and not-yet-treated controls. We use version 2.5.1 of `did`, with its default `faster_mode = TRUE`. This version is much faster than earlier ones: in a previous version of this note, based on version 2.1.2, `did` took about 28 minutes on the large dataset, against less than a minute now. Since `did` lets covariate effects differ by cohort and period, and always computes influence-function standard errors, it is comparable to our specification with covariates interacted with cohorts and corrected standard errors (last column).

Table 2: Median execution time on a medium dataset (100,000 observations, wide format)

Covariates interacted with cohort	No		Yes	
Standard errors	Usual	Corrected	Usual	Corrected
`lpdid`	0.78s	n.a.	–	n.a.
`lpdid_rw`	1.04s	1.46s	–	–
`lpcsa`	1.80s	n.a.	–	n.a.
`lpdid_adj`	1.41s	2.08s	1.62s	3.85s
`lpdid_ipw`	5.08s	5.57s	5.30s	8.80s
`lpcsa_ipw`	5.11s	n.a.	6.03s	n.a.
`lpdid_dr`	6.38s	8.44s	6.96s	11.44s
`did`, `est_method="reg"`				0.56s
`did`, `est_method="ipw"`				1.23s
`did`, `est_method="dr"`				1.39s

Median of 10 iterations. Data preparation (`lpdidcsa_data`, wide format) takes 1.60s and produces a 29.4 MB dataset with 41 columns. Covariates interacted with cohort: `time_x_controls = TRUE`. n.a.: no correction available for this method. –: not run. `did` uses `faster_mode = TRUE`.

The estimators without first step are the fastest: `lpdid` and `lpdid_rw` take about 10 seconds on 4 million observations, and `lpcsa`, which estimates every cohort effect, about twice as long. Regression adjustment remains fast, since it only adds a second linear regression: without interactions and corrections, it is about as fast as the regression estimator of `did` (21s versus 19s). Methods using a propensity score are slower, because they estimate one logit per horizon on the whole clean sample. Standard-error corrections and interactions between

Table 3: Median execution time on a medium dataset, long format

Method	One regression per horizon	Single regression
`lpdid`	1.32s	5.73s
`lpdid_rw`	2.04s	4.21s
`lpcsa`	2.23s	32.61s
`lpdid_adj`	2.75s	n.a.
`lpdid_ipw`	5.22s	12.60s
`lpcsa_ipw`	4.73s	37.59s
`lpdid_dr`	7.53s	n.a.

Median of 10 iterations (one regression per horizon) and 5 iterations (single regression). Covariates not interacted with cohort. Corrected standard errors in the first column for `lpdid_rw`, `lpdid_adj`, `lpdid_ipw` and `lpdid_dr`; the correction is not available for the single regression. n.a.: `lpdid_adj` and `lpdid_dr` are not available with a single regression. Data preparation (long format) takes 0.46s and produces a 150.6 MB dataset with 1,795,000 rows and 14 columns.

Table 4: Median execution time on a large dataset (4,000,000 observations, wide format)

Covariates interacted with cohort	No		Yes	
Standard errors	Usual	Corrected	Usual	Corrected
`lpdid`	9.0s	n.a.	–	n.a.
`lpdid_rw`	12.7s	27.7s	–	–
`lpcsa`	24.0s	n.a.	–	n.a.
`lpdid_adj`	20.8s	52.5s	29.9s	82.8s
`lpdid_ipw`	111.6s	159.6s	172.8s	253.8s
`lpcsa_ipw`	123.0s	n.a.	185.4s	n.a.
`lpdid_dr`	147.6s	210.6s	241.2s	379.8s
`did`, `est_method="reg"`				18.7s
`did`, `est_method="ipw"`				39.8s
`did`, `est_method="dr"`				43.5s

Median of 5 iterations. Data preparation (`lpdidcsa_data`, wide format) takes 54.3s and produces a 1,175 MB dataset with 41 columns. Same conventions as Table 2.

covariates and cohorts both add to the run time; together, they multiply it by 1.5 to 4.

For covariate-adjusted estimators with comparable specifications, the `did` package is faster, by a factor of about 4 (RA) to 9 (DR). Data preparation also adds a fixed cost to our approach. The advantages of `lpdidcsa` therefore lie less in speed than in the flexibility described in Section 6: pair balancing, non-absorbing treatments, high-dimensional fixed effects, and a simple regression representation.

Appendices: standard-error corrections

The appendices use the following notation. We consider one horizon h and its clean sample, indexed by $i = 1, \ldots, N$. $D_i = \Delta T_{t,i}$ is the treatment dummy, $\Delta Y_i = \Delta^h y_{t,i}$ the long difference, $t(i)$ the reference date of observation i, b_i its survey weight (1 if none), and $g(i)$ its cluster. For a vector of first-step parameters θ, estimated by solving $\sum_i s_i(\theta) = 0$, we write ℓ_i its *linear representation*: $\hat{\theta} - \theta \approx \sum_i \ell_i$, with $\ell_i = H^{-1} s_i$ and H the (minus) Hessian of the first-step objective.

A Regression adjustment

The outcome model is a weighted regression of ΔY_i on regressors Z_i (date dummies and covariates) on the controls. Let $\hat{\theta}$ be its coefficients, $s_i = b_i Z_i (\Delta Y_i - Z_i' \hat{\theta})$ the score of control i, and $H_0 = \sum_{i:D_i=0} b_i Z_i Z_i'$. The estimate is

$$\hat{\beta}_{RA} = \frac{1}{N_1} \sum_{i:D_i=1} b_i \, r_i, \qquad r_i = \Delta Y_i - Z_i' \hat{\theta}, \qquad N_1 = \sum_{i:D_i=1} b_i.$$

Uncorrected standard error. The package regresses the residuals r_i of the treated on a constant, with clustered standard errors. This treats $\hat{\theta}$ as known.

Corrected standard error. The estimate depends on $\hat{\theta}$ through the predictions. Its derivative with respect to θ is $-\bar{Z}_1$, with $\bar{Z}_1 = \sum_{i:D_i=1} b_i Z_i / N_1$ the average regressors of the treated. The contribution of each observation is thus

$$\phi_i = \begin{cases} b_i (r_i - \hat{\beta}_{RA}) / N_1 & \text{if } D_i = 1 \text{ (own residual)}, \\ -\bar{Z}_1' H_0^{-1} s_i & \text{if } D_i = 0 \text{ (effect through } \hat{\theta}). \end{cases}$$

The variance is $V = \sum_g \left(\sum_{i:g(i)=g} \phi_i \right)^2$. Summing within clusters before squaring captures the covariance between the two terms when a cluster contains both treated and controls. This formula reproduces exactly the standard error of the Stata `lpdid` command (`listreg` implementation).

B Doubly robust estimation

Let $\hat{\mu}_{0,i} = Z_i' \hat{\theta}$ be the prediction of the outcome model, and $\hat{p}_i$ the propensity score, from a logit with regressors W_i and coefficients $\hat{\gamma}$. With $\tilde{b}_i$ the survey weights normalized to mean one, τ the trimming threshold and $r_i = \Delta Y_i - \hat{\mu}_{0,i}$, define

$$w_{1,i} = D_i \tilde{b}_i, \qquad w_{0,i} = (1 - D_i)\, \mathbb{1}[\hat{p}_i < \tau]\, \tilde{b}_i \frac{\hat{p}_i}{1 - \hat{p}_i},$$

$$\hat{\eta}_1 = \frac{\sum_i w_{1,i}\, r_i}{\sum_i w_{1,i}}, \qquad \hat{\eta}_0 = \frac{\sum_i w_{0,i}\, r_i}{\sum_i w_{0,i}}, \qquad \hat{\beta}_{DR} = \hat{\eta}_1 - \hat{\eta}_0.$$

Uncorrected standard error. The package computes the clustered standard error of the mean of $w_{1,i} r_i / \bar{w}_1 - w_{0,i} r_i / \bar{w}_0$, where $\bar{w}_1$ and $\bar{w}_0$ are sample means. This treats $\hat{\theta}$ and $\hat{\gamma}$ as known.

Corrected standard error. Following Sant'Anna and Zhao (2020), the influence function of observation i is

$$\psi_i = \underbrace{\frac{w_{1,i}(r_i - \hat{\eta}_1)}{\bar{w}_1} - \frac{w_{0,i}(r_i - \hat{\eta}_0)}{\bar{w}_0}}_{\text{own contribution}} - \underbrace{\ell_i^{\theta\prime} \left(\bar{Z}_1 - \bar{Z}_0^w \right)}_{\text{outcome model}} - \underbrace{\frac{\ell_i^{\gamma\prime} M}{\bar{w}_0}}_{\text{propensity score}},$$

where $\bar{Z}_1$ is the average of Z among the treated, $\bar{Z}_0^w = \sum_i w_{0,i} Z_i / \sum_i w_{0,i}$ its weighted average among the controls, $M = \frac{1}{N} \sum_i w_{0,i}(r_i - \hat{\eta}_0) W_i$, and ℓ_i^θ and ℓ_i^γ are the linear representations of the outcome model (zero for treated observations) and of the logit. The outcome-model term vanishes when the weighted controls have the same average regressors as the treated. The standard error is the clustered standard error of the mean of ψ_i. Trimming is treated as fixed.

C Reweighted and IPW LP-DiD

Both estimators run a weighted regression of ΔY_i on D_i and date fixed effects (and possibly controls), with observation weights

$$w_i = \frac{b_i}{1 - q_{t(i)}}, \qquad q_t = \frac{\sum_{i:t(i)=t} b_i D_i}{B_t}, \qquad B_t = \sum_{i:t(i)=t} b_i,$$

where b_i is the survey weight for `lpdid_rw`, and the survey weight times the IPW weight (1 for treated, $\hat{p}_i/(1-\hat{p}_i) = e^{W_i'\hat{\gamma}}$ for controls) for `lpdid_ipw`. Let s_i be the score of observation i in this regression and H its Hessian. The treatment coefficient satisfies $\hat{\beta} - \beta \approx e'H^{-1}\sum_i s_i$, where e selects the treatment coefficient.

Uncorrected standard error. The usual clustered standard error treats the weights w_i as known: $V_u = \sum_g \left(e'H^{-1}\sum_{i:g(i)=g} s_i\right)^2$.

Correction for the treated shares. The weights depend on the estimated shares q_t. Changing q_t changes the score of each observation of date t by $s_i/(1-q_t)$, so that the total effect is $G_t = \sum_{i:t(i)=t} s_i/(1-q_t)$.[3] Since $\hat{q}_t - q_t \approx \sum_{j:t(j)=t} b_j(D_j - q_t)/B_t$, the corrected contribution of observation i is

$$\phi_i = s_i + G_{t(i)}\frac{b_i(D_i - q_{t(i)})}{B_{t(i)}}.$$

Additional correction for the propensity score (IPW). The control weights also depend on $\hat{\gamma}$: directly, through the odds, and indirectly, through the shares q_t. The derivative of the total score with respect to γ is

$$J = \sum_i s_i(1 - D_i)\, W_i' + \sum_t G_t \frac{\partial q_t}{\partial \gamma'}, \qquad \frac{\partial q_t}{\partial \gamma} = -\frac{q_t}{B_t}\sum_{i:t(i)=t,\, D_i=0} b_i W_i.$$

Each observation j of the propensity-score sample then adds $J\,\ell_j^{\gamma}$ to the contributions of its cluster.

Reported standard error. The corrected variance is

$$V_c = \sum_g \left(e'H^{-1}\sum_{j:g(j)=g}\phi_j\right)^2,$$

including the propensity-score terms for IPW. To keep the small-sample adjustment of `fixest`, the package reports `se_correct` = `se` $\times \sqrt{V_c/V_u}$. Trimming is treated as fixed.

[3] The effects through the date fixed effects vanish by their first-order conditions.

References

Abadie, Alberto (2005). Semiparametric difference-in-differences estimators. *The Review of Economic Studies 72*(1), 1–19.

Bergé, Laurent (2018). Efficient estimation of maximum likelihood models with multiple fixed-effects: the R package FENmlm. CREA Discussion Paper 13, CREA, University of Luxembourg.

Callaway, Brantly and Pedro H. C. Sant'Anna (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics 225*(2), 200–230.

Dube, Arindrajit, Daniele Girardi, Òscar Jordà, and Alan M. Taylor (2025). A local projections approach to difference-in-differences. *Journal of Applied Econometrics 40*(7), 741–758.

Sant'Anna, Pedro H. C. and Jun Zhao (2020). Doubly robust difference-in-differences estimators. *Journal of Econometrics 219*(1), 101–122.

Sun, Liyang and Sarah Abraham (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics 225*(2), 175–199.